

BCE-C514

SEMESTER EXAMINATION-2022

B.TECH. - V SEMESTER

Paper Title : CLOUD COMPUTING

206301076

Time: 3 hours

Max. Marks: 70

Note: Question Paper is divided into two sections: A and B. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any five questions in about 150 words each. Each question carries six marks.
(5 × 6 = 30 Marks)

Question-1: Define Cloud Computing. What are the characteristics of cloud computing?

Question-2: What are different cloud services?

Question-3: Define Virtualization. What are the Pros and cons of Virtualization?

Question-4: Discuss the following concepts in the context of cloud computing:

- a) Billing and Metering Services
- b) Tooling and Automation.

Question-5: Write different types of internal security breaches in cloud computing. Also, explain the steps to reduce cloud security breaches.

Question-6: What is the role of hypervisor? What are different types of hypervisors?

Question-7: List out and brief the features of Amazon Simple Queue Service.

Question-8: Describe the storage structure of S3 bucket with neat sketch and write rules in order to make the bucket available for public access.

Question-9: What are the major features of Aneka application model?

Question-10: What is AWS? What types of services does it provide?

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any **four** questions in detail. Each question carries 10 marks.
(4 × 10 = 40 Marks)

- Question-1: Explain the cloud deployment models in detail. Also, outline their benefits and limitations while implementing and application.
- Question-2: Discuss, in detail, the cloud security reference model and explain how cloud security is integrated into the design of application.
- Question-3: Describe the architecture of Windows Azure. Explain the various cloud services provided by Windows Azure.
- Question-4: Explain the differences between full-virtualization and para-virtualization and give one example of VMM (virtual machine monitor), that was built in each of the two categories.
- Question-5: Describe the Google App Engine Platform Architecture with suitable diagram.
- Question-6: Define Multi Tenancy. Discuss the different techniques used for securing multi-tenant environments.
- Question-7: What are the different types of monitoring agents to monitor cloud usage? Discuss the working of each using suitable block diagram.
- Question-8: Explain techniques for the risks assessment and management for cloud.